

The Solow Growth Model with Land and Natural Resources

The question and assumptions

How do a fixed factor and a non-renewable resource affect long-run growth?

$$\frac{\dot{L}_t}{L_t} = n, \quad \frac{\dot{A}_t}{A_t} = x, \quad \frac{\dot{R}_t}{R_t} = -b, \quad \dot{T}_t = 0.$$

- ▶ L_t : labour, growing at rate n .
- ▶ A_t : labour-augmenting technology, growing at rate x .
- ▶ R_t : resource input, declining at rate $b \geq 0$.
- ▶ T_t : land, fixed over time.

Production and capital accumulation

$$Y_t = (A_t L_t)^{1-\alpha-\beta-\gamma} K_t^\alpha R_t^\beta T_t^\gamma. \quad (1)$$

Assume $0 < \alpha < 1$, $\beta, \gamma \geq 0$, and $\alpha + \beta + \gamma < 1$.

$$\dot{K}_t = I_t - \delta K_t = sY_t - \delta K_t, \quad 0 < s < 1.$$

Dividing by K_t gives

$$\frac{\dot{K}_t}{K_t} = s \frac{Y_t}{K_t} - \delta. \quad (2)$$

Capital growth depends on the output–capital ratio.

The growth rate of output

Take logs of the production function:

$$\ln Y_t = (1 - \alpha - \beta - \gamma)(\ln A_t + \ln L_t) + \alpha \ln K_t + \beta \ln R_t + \gamma \ln T_t.$$

Differentiate with respect to time:

$$\frac{\dot{Y}_t}{Y_t} = (1 - \alpha - \beta - \gamma) \left(\frac{\dot{A}_t}{A_t} + \frac{\dot{L}_t}{L_t} \right) + \alpha \frac{\dot{K}_t}{K_t} + \beta \frac{\dot{R}_t}{R_t} + \gamma \frac{\dot{T}_t}{T_t}.$$

Substitute the assumed growth rates:

$$\frac{\dot{Y}_t}{Y_t} = (1 - \alpha - \beta - \gamma)(x + n) - \beta b + \alpha \frac{\dot{K}_t}{K_t}. \quad (3)$$

The balanced growth path

Suppose output and capital grow at the same constant rate g :

$$\frac{\dot{Y}_t}{Y_t} = \frac{\dot{K}_t}{K_t} = g.$$

Substituting into (3):

$$\begin{aligned} g &= (1 - \alpha - \beta - \gamma)(x + n) - \beta b + \alpha g, \\ (1 - \alpha)g &= (1 - \alpha - \beta - \gamma)(x + n) - \beta b. \end{aligned}$$

Hence

$$\boxed{g = \frac{1 - \alpha - \beta - \gamma}{1 - \alpha}(x + n) - \frac{\beta}{1 - \alpha}b.} \quad (4)$$

The output–capital ratio on the balanced growth path

From capital accumulation, on the balanced growth path:

$$g = s \left(\frac{Y}{K} \right)^* - \delta.$$

Thus

$$\boxed{\left(\frac{Y}{K} \right)^* = \frac{g + \delta}{s}.$$

- ▶ A positive ratio requires $g + \delta > 0$.
- ▶ The convergence argument below assumes this condition holds.
- ▶ On this path, Y_t and K_t grow at rate g , so Y_t/K_t stays constant.

Convergence: the key relationship

Subtract the balanced-growth relation from (3):

$$\begin{aligned}\frac{\dot{Y}_t}{Y_t} - g &= (1 - \alpha - \beta - \gamma)(x + n) - \beta b + \alpha \frac{\dot{K}_t}{K_t} - g \\ &= \alpha \left(\frac{\dot{K}_t}{K_t} - g \right).\end{aligned}$$

Therefore

$$\boxed{\frac{\dot{Y}_t}{Y_t} - g = \alpha \left(\frac{\dot{K}_t}{K_t} - g \right).} \quad (5)$$

Since $0 < \alpha < 1$, output growth is closer to g than capital growth is.

Also,

$$\frac{d}{dt} \ln \left(\frac{Y_t}{K_t} \right) = \frac{\dot{Y}_t}{Y_t} - \frac{\dot{K}_t}{K_t}.$$

Case (i): capital growth exceeds g

Suppose $\frac{\dot{K}_t}{K_t} > g$. From (5):

$$0 < \frac{\dot{Y}_t}{Y_t} - g = \alpha \left(\frac{\dot{K}_t}{K_t} - g \right) < \frac{\dot{K}_t}{K_t} - g.$$

Hence

$$g < \frac{\dot{Y}_t}{Y_t} < \frac{\dot{K}_t}{K_t}.$$

- ▶ Output grows more slowly than capital, so Y_t/K_t falls.
- ▶ From (2), $\frac{\dot{K}_t}{K_t} = sY_t/K_t - \delta$ also falls.
- ▶ Capital growth moves down toward g .

Case (ii): capital growth is below g

Suppose $\frac{\dot{K}_t}{K_t} < g$. From (5):

$$\frac{\dot{K}_t}{K_t} - g < \frac{\dot{Y}_t}{Y_t} - g = \alpha \left(\frac{\dot{K}_t}{K_t} - g \right) < 0.$$

Hence

$$\frac{\dot{K}_t}{K_t} < \frac{\dot{Y}_t}{Y_t} < g.$$

- ▶ Output grows more quickly than capital, so Y_t/K_t rises.
- ▶ From (2), $\frac{\dot{K}_t}{K_t} = sY_t/K_t - \delta$ also rises.
- ▶ Capital growth moves up toward g .

Case (iii): capital growth equals g

Suppose $\frac{\dot{K}_t}{K_t} = g$. From (5):

$$\frac{\dot{Y}_t}{Y_t} - g = \alpha(g - g) = 0, \quad \frac{\dot{Y}_t}{Y_t} = g.$$

Therefore

$$\frac{d}{dt} \ln \left(\frac{Y_t}{K_t} \right) = g - g = 0.$$

- ▶ The output–capital ratio remains constant.
- ▶ Equation (2) keeps capital growth equal to g .
- ▶ Once the economy is on the balanced growth path, it stays there.

For positive initial inputs and $g + \delta > 0$, the economy converges toward this path.

Long-run growth per capita

Since labour grows at rate n , output per worker grows at rate $g - n$.

$$\begin{aligned} g - n &= \frac{1 - \alpha - \beta - \gamma}{1 - \alpha} (x + n) - \frac{\beta}{1 - \alpha} b - n \\ &= \frac{1 - \alpha - \beta - \gamma}{1 - \alpha} x + \left(\frac{1 - \alpha - \beta - \gamma}{1 - \alpha} - 1 \right) n - \frac{\beta}{1 - \alpha} b. \end{aligned}$$

Thus

$$g - n = \underbrace{\frac{1 - \alpha - \beta - \gamma}{1 - \alpha} x}_{(a)} - \underbrace{\frac{\beta}{1 - \alpha} b}_{(b)} - \underbrace{\frac{\beta + \gamma}{1 - \alpha} n}_{(c)}. \quad (6)$$

Capital per worker and consumption per worker also grow at $g - n$, since $C_t = (1 - s)Y_t$.

Interpretation of long-run growth per capita

$$g - n = \frac{1 - \alpha - \beta - \gamma}{1 - \alpha}x - \frac{\beta}{1 - \alpha}b - \frac{\beta + \gamma}{1 - \alpha}n.$$

- (a) **Technological progress:** sufficiently rapid growth in A_t can sustain positive growth per capita.
- (b) **Resource depletion:** less resource input is available each period, reducing production growth.
- (c) **Population growth:** more workers share fixed land and a declining resource input. This congestion reduces growth per capita.

For $b, n \geq 0$, positive growth per capita requires

$$(1 - \alpha - \beta - \gamma)x > \beta b + (\beta + \gamma)n.$$